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To the Editors, *SIAM Journal on Mathematics of Data Science*

Dear Editors,

Please consider the enclosed manuscript, “Keep the Angle: A Geometry-Preserving Basis in Spectral Embeddings,” for publication in *SIMODS*.

Summary. The commute-time (resistance) embedding built from the low eigenvectors of a graph Laplacian degenerates on large graphs (von Luxburg–Radl–Hein). We show the geometry does not vanish but is carried by the *angular* coordinate—the row-normalization of Ng–Jordan–Weiss spectral clustering, here given a metric-geometric rather than a cluster-separation reading. The contribution is organized as four claims: (1) an exact normalization identity locating the mechanism in *tangential noncollapse*, and a deterministic angular-transfer theorem; (2) a conditional graph-to-continuum transfer, carried by the random-walk eigenvectors and the sup-norm eigenvector rates of Calder–García Trillos–Lewicka, together with an *unconditional* bi-Lipschitz statement on the flat torus; (3) a filter dichotomy—uniform-in-truncation for the heat filter, scale-dependent for the commute filter, with an exact rank-collapse theorem for the plain filter on the circle; and (4) a controlled empirical study across five substrate families. We believe the topic—when and why row-normalized spectral embeddings recover intrinsic geodesic order—is squarely within the journal’s scope.

Originality and exclusivity. This manuscript is original, has not been published, and is not under review at any other journal or conference. We intend to post it as a preprint, consistent with SIAM policy.

Related work by the author (overlap disclosure). In the interest of full transparency we note two of the author’s own companion artifacts that the manuscript cites for context, and which are *not* overlapping submissions:

- *Observation Theory: geometry, distortion, and reliability as properties of observation* (technical report / preprint, Zenodo 10.5281/zenodo.21423315). This is a broader, consumer-relative framework; the present paper uses only its “tangential-noncollapse (A2)” vocabulary as an organizing label and shares no theorem with it.
- *TurboQuant-Pro* (open-source software and accompanying technical note). Cited once, as a concrete boundary example (per-vector angular quantization of attention *keys* fails because scale is signal there). It is a software project, not a journal submission.

Neither is submitted or accepted elsewhere; copies can be provided to the editors and referees on request.

Revision note. The manuscript underwent detailed pre-submission review; the principal changes are summarized in the point-by-point response below, and the two new results (the S^1 rank-collapse theorem and the random-walk transfer repair) were subjected to an independent cold re-derivation before submission.

Thank you for your consideration.

Sincerely,

Andrew H. Bond

Response to review

We are grateful for the careful reading. Below we address each major point; the verdict was *major revision*, and every item is now resolved. Section and label references are to the revised manuscript.

1. Omitted prior use of the normalization. We now cite Sharma–Heraud–Mateus (2012), who apply unit-hypersphere normalization directly to the commute-time embedding for shape registration, and we reframe the novelty accordingly: the normalization is not new; the metric-geometric account of *when its angular distances recover intrinsic geodesic order* is our contribution (§1).

2. Proposition 14 did not prove rank collapse. Correct—distance concentration does not imply rank collapse. We split the result into an honest general-manifold *concentration* proposition and a new *exact* theorem (Theorem, `thm:s1collapse`) proving that on S^1 both Spearman and Kendall correlation with geodesic distance vanish, via the Dirichlet kernel and two-scale equidistribution. This is materially stronger than weakening the claim.

3. Theorem 8 normalization gap (L_{sym} vs. L_{rw}). Adopted the suggested repair. The angular map is invariant to the exact per-node factor $\sqrt{d_i}$ ($\Psi_i^{\text{sym}} = \sqrt{d_i} \Psi_i^{\text{rw}}$), so we carry the sup-norm transfer on the random-walk eigenvectors—for which the cited rates are stated—and read off the algorithm’s L_{sym} angle by exact cancellation. No degree-concentration hypothesis is needed; the theorem now holds under non-uniform sampling. Step 3 is written as an external lemma (`lem:transfer`) composing named rate theorems (Calder–García Trillos for eigenvalue/gap stability; Calder–García Trillos–Lewicka for the L^∞ and $C^{0,1}$ eigenvector rates), making the theorem unconditional in m at each fixed m ; the m -dependence of those constants is stated explicitly as the bridge to the open problem.

4. The central conjecture was quantified too broadly. Conjecture is now scoped to an admissible mode schedule (multiplet-closed m at a spectral gap, $m \geq m_0(M)$, the spectral-window condition for growing m , pairs above the resolution scale), with “uniform” meaning a bound over that family. We name the hypothesis (H1)-uniform, prove its two endpoints (unconditional on the torus; a continuum theorem for the heat filter, with the commute filter shown *not* m -uniform at fixed scale), and state the single sharp open problem—a gap-uniform lower bound on the smallest singular value of the truncated-eigenmap Jacobian (`conj:sgmamin`)—whose resolution closes the metric form.

5. Sliding between two embeddings; $d \geq 3$ vs $d \geq 2$. We now state the radial claim at three fixed levels and never mix them: proved for the full unnormalized embedding at $d \geq 3$; empirical for the truncated normalized radius the algorithm computes ($\rho \approx 0.79$ – 0.87 vs. 0.98 for the full radius); borderline at $d = 2$. A “Status of the radial claim” block records that the gap between the proved and empirical levels is exactly the truncation, which no theorem here bridges.

6. Proposition 13 remainder. We supply the required differentiated pointwise Weyl law as a displayed lemma (`lem:dweyl`), citing Hörmander for the on-diagonal projector and Canzani–Hanin for the mixed-derivative remainder (both already in the bibliography), and add a cutoff-insensitivity remark noting the S^1 case needs no Weyl input.

7. The “scaling law” is not a law. Retitled to “an exploratory dimension–fidelity trend.” The five rule-level summaries are now the primary table; the twenty-point fit is presented as a descriptive trend line with its rule-clustered CI, and Proposition (rank) is repositioned as *consistent with, and not deriving*, the trend.

8. Methods too compressed; graph vs. manifold; replication. Added a per-section protocol

table; we report both the measured dimension d_G and the ground-truth d_M where it exists, noting the finite-sample estimator bias shifts the axis but not the ordering; and the headline uncertainty is reported over independent graph draws (pair-resampling intervals are labeled as optimistic floors).

Additional comments. Observer-theory and Wolfram framing are removed from the title and abstract and confined to a single discussion paragraph; the abstract is rewritten to ≤ 250 words with references spelled out; terminology is softened from “basis” to “representation” where literally apt; the figure caption/text discrepancies are reconciled and min–max ranges are labeled as ranges rather than confidence intervals.